



THE SCOTS COLLEGE

Year 12 Mathematics Extension 2 Assessment 1 February 2007

GENERAL INSTRUCTIONS

- Working time - 50 minutes
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question

TOTAL MARKS: 40

WEIGHTING: 10%

- Attempt all questions

OUTCOMES

QUESTION/ OUTCOME	A student combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions	A student uses the relationship between algebraic and geometric representations of complex numbers
Question 1	/10	
Question 2		/30
TOTAL	/40	
PERCENTAGE		

QUESTION 1 [10 MARKS]**MARKS**

- (i) Sketch the curve $f(x) = x^2 - 2x - 3$ on the number plane provided at back. [2]
- (ii) Sketch $y = \frac{1}{f(x)}$ on a separate diagram [2]
- (iii) Sketch $y = \sqrt{f(x)}$ on a separate number plane. [2]
- (iv) Sketch $y = f(|x|)$ on a separate diagram [2]
- (v) Sketch $\ln f(x)$ on the number plane displaying the graph of $y = f(x)$. [2]

QUESTION 2 [10 MARKS]

The points A and B on the Argand plane provided represent the complex numbers z_1 and z_2 respectively.

Mark the position of the following complex numbers with the letter indicated, on the number plane provided at back:

- (i) $C = z_1 + z_2$ [2]
- (ii) $D = z_1 - z_2$ [2]
- (iii) $E = iz_2$ [2]
- (iv) $F = z_1 z_2$ [2]
- (v) $G = \overline{z_2 - z_1}$

QUESTION 3 [10 MARKS]**MARKS**

Sketch the graphs of the following on the Argand plane provided.

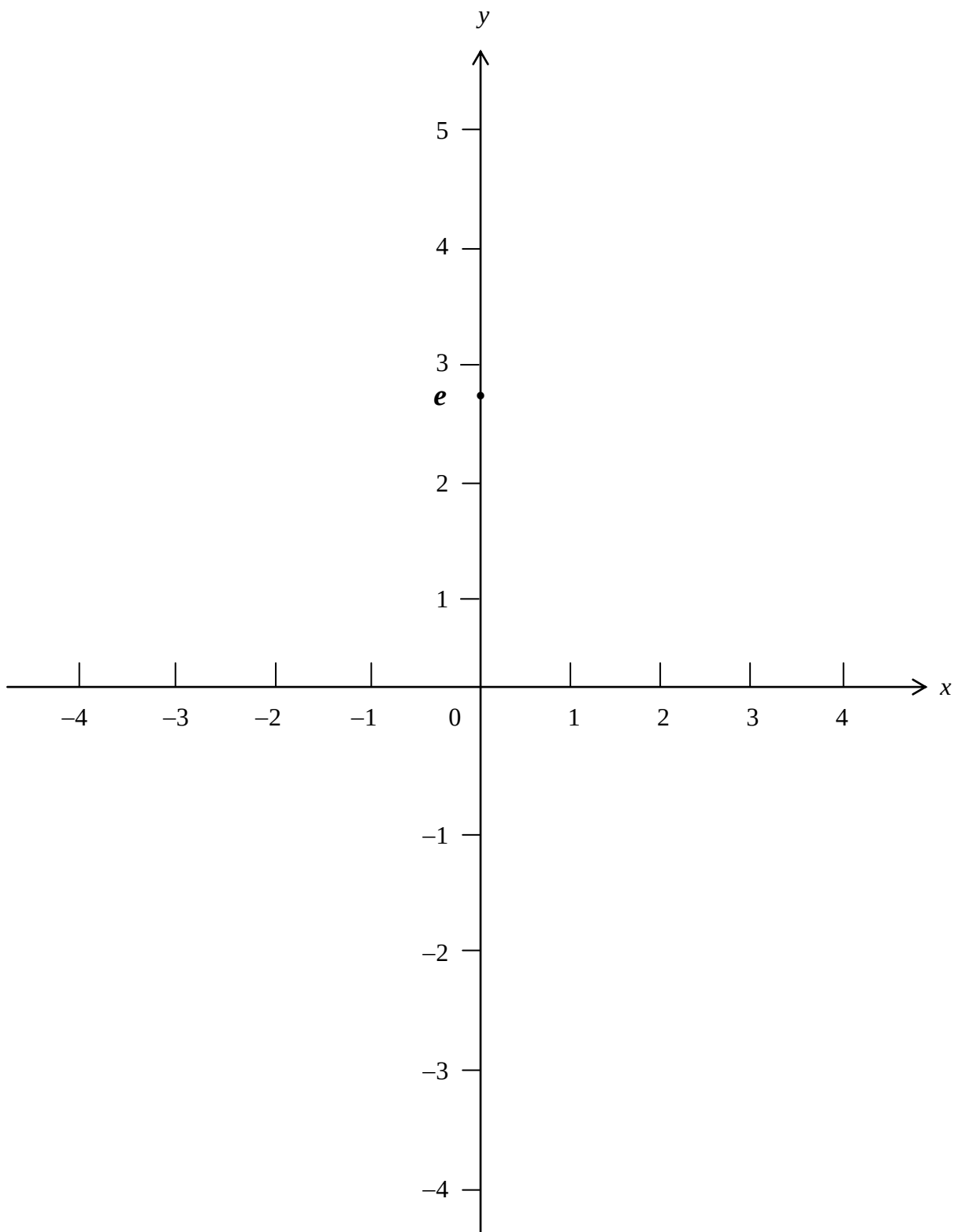
- a. (i) $0 \leq \operatorname{Im}|z| \leq \sqrt{3}, 1 \leq \operatorname{Re}(z) \leq 3$ [2]
- (ii) Determine all possible values of $\arg(z)$. [1]
- b. (i) Sketch $|z - 2 + 2i| \leq 2$ [2]
- (ii) Hence or otherwise determine the maximum value of $|z|$. [2]
- c. (i) Sketch the locus represented by $\operatorname{Arg}\left(\frac{z-1}{z-i}\right) = \frac{\pi}{4}$ [2]
- (ii) Describe this locus geometrically. [1]

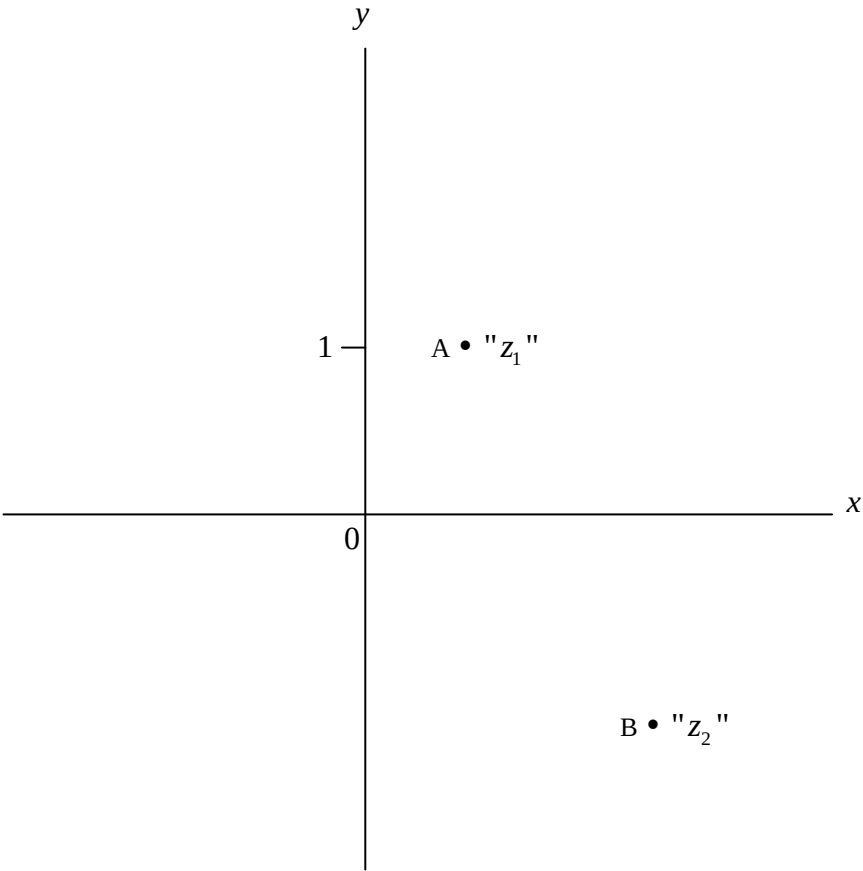
QUESTION 4 [10 MARKS]**MARKS**

- a. (i) Express $z = 1 + \sqrt{3}i$ on the form $R \operatorname{cis} \theta$. [2]
- (ii) Hence verify the result $|z|^2 = z\bar{z}$ [1]
- (iii) Find the square roots of z . [2]
- b. (i) Find the five fifth roots of unity. [3]
- (ii) By considering $z^5 = 1$ show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ [2]

QUESTION 1 ANSWER SHEET

Parts (i) and (v) ONLY

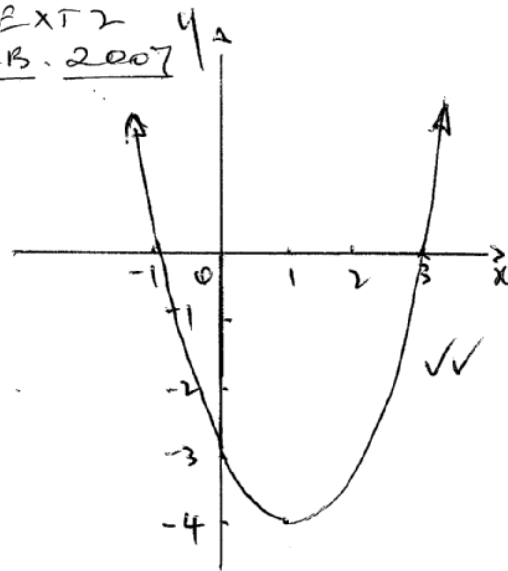




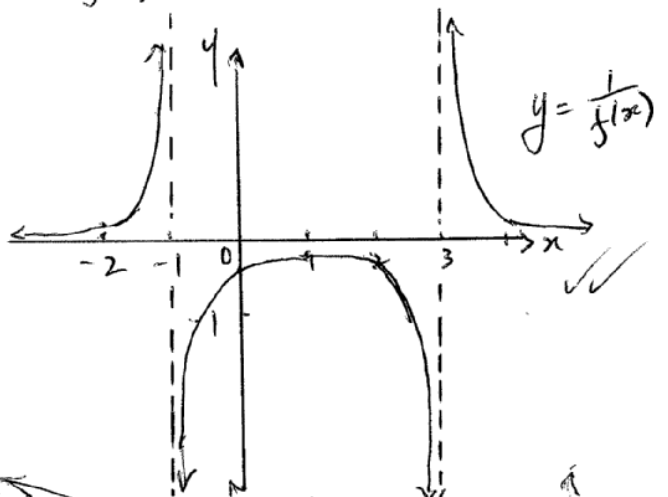
Q1. (i) $f(x) = x^2 - 2x - 3$
 $= (x-3)(x+1)$.

$f(0) = -3$

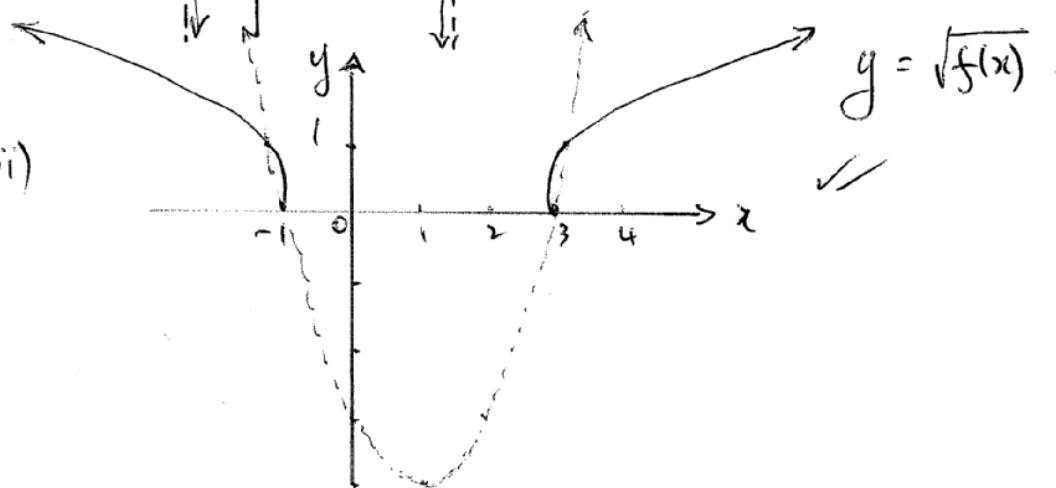
$f(x) = 0$ where $x = -1$ or 3 .



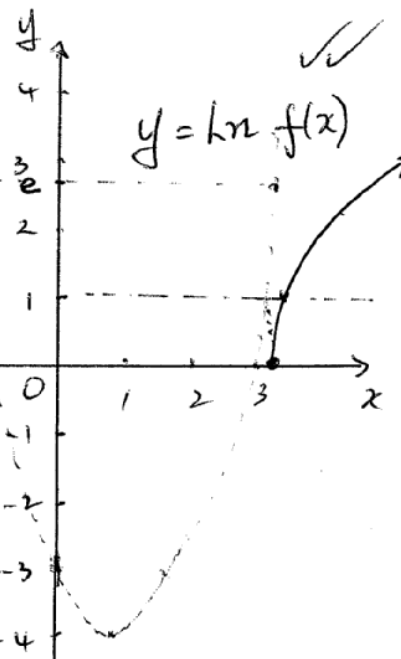
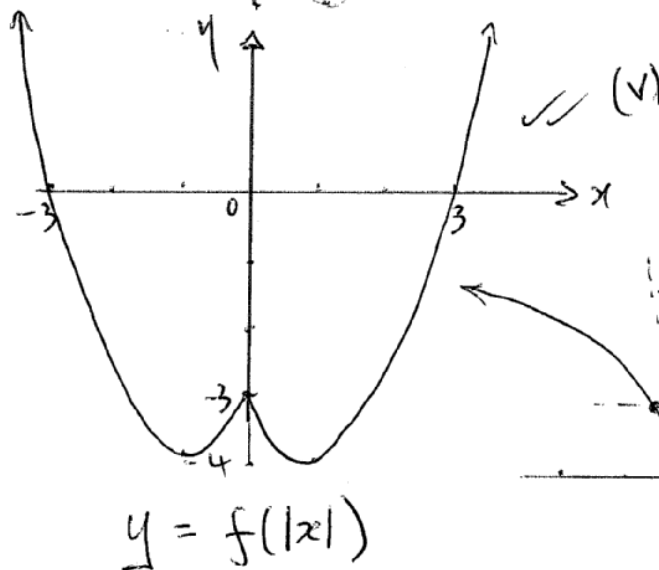
(ii)



(iii)

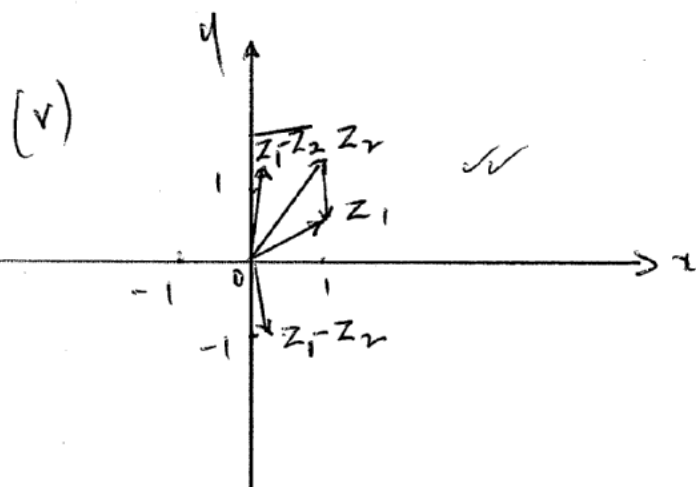
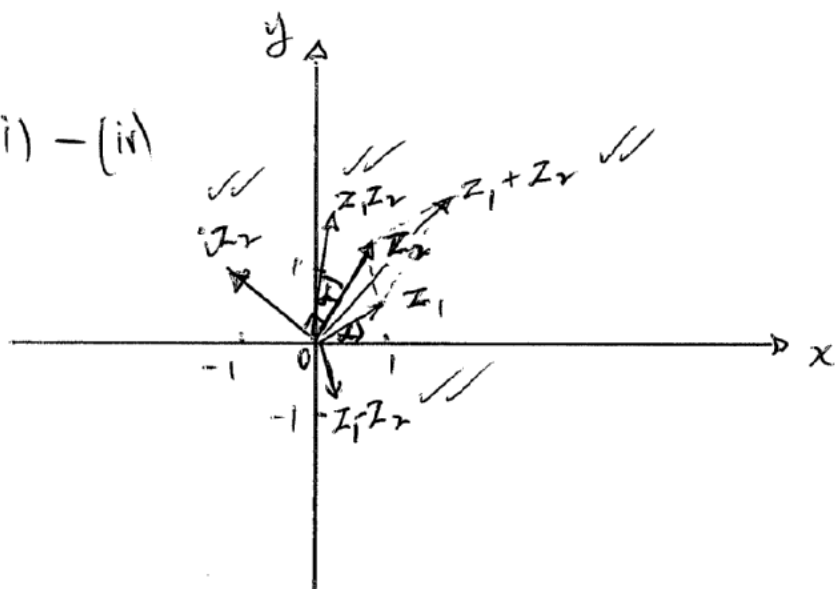


(iv)

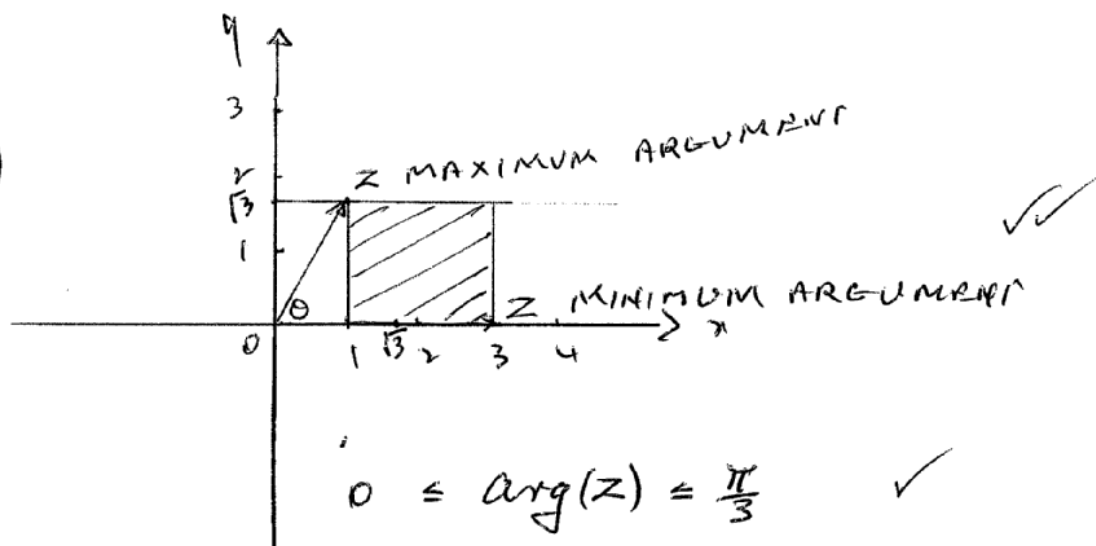


Q2.

(i) - (iv)



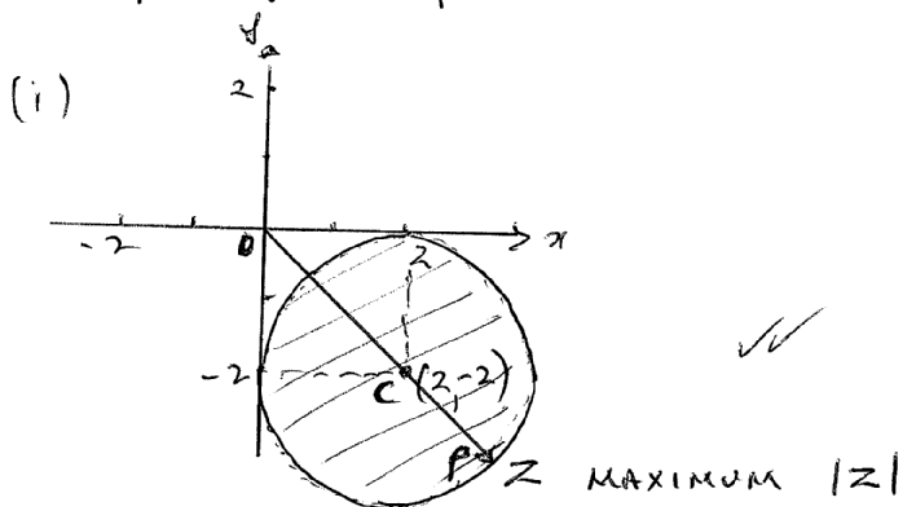
Q3(a)(i)



(ii)

$$0 \leq \arg(z) \leq \frac{\pi}{3}$$

(b) $|z - (2 - 2i)| \leq 2$



(ii)

$$OC = 2\sqrt{2}$$

$$CP = 2 \text{ (Radius)}$$

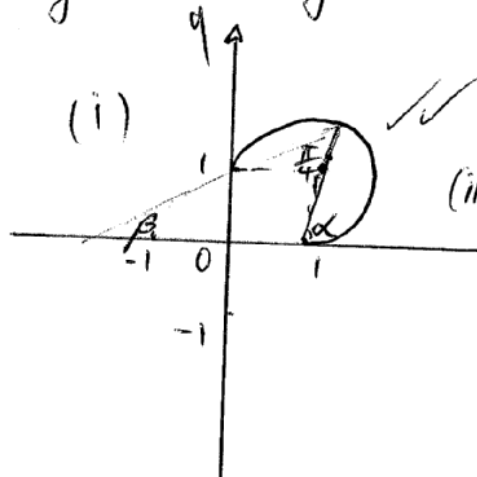
So $OP = 2 + 2\sqrt{2}$

$$|z| = 2(1 + \sqrt{2})$$

(c) $\arg(z-1) - \arg(z-i) = \frac{\pi}{4} \rightarrow$ let $\alpha = \arg(z-1)$

and $\beta = \arg(z-i)$

So $\alpha - \beta = \frac{\pi}{4}$



(ii) Major arc of a circle
centre $C(1,1)$ and
radius 1 unit terminating
at $(0,1)$ and $(1,0)$.

Q4 (a) (i)

$$Z = 1 + \sqrt{3}i$$

$$= 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$Z = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

(ii)

$$|Z| = 2$$

$$|Z|^2 = 4$$

$$Z \bar{Z} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \cdot 2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$$

$$= 4 \left(\cos^2 \frac{\pi}{3} - i^2 \sin^2 \frac{\pi}{3} \right)$$

$$= 4 \left(\cos^2 \frac{\pi}{3} + \sin^2 \frac{\pi}{3} \right)$$

$$= 4$$

$$= |Z|^2$$

(iii)

$$\text{For } Z^2 = 1 + \sqrt{3}i$$

$$|Z|^2 = |1 + \sqrt{3}i| = 2$$

$$|Z| = \sqrt{2} \text{ and so } Z = \sqrt{2} \operatorname{cis} \theta$$

$$\text{Now } (\sqrt{2} \operatorname{cis} \theta)^2 = 2 \operatorname{cis} \frac{\pi}{3}$$

$$2 \operatorname{cis} 2\theta = 2 \operatorname{cis} \frac{\pi}{3}$$

$$\cos 2\theta + i \sin 2\theta = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\cos 2\theta = \frac{1}{2}, \sin 2\theta = \frac{\sqrt{3}}{2}$$

$$2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \dots, 2\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3} \dots$$

$$\text{So } 2\theta = \frac{\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{7\pi}{6}$$

$$\text{and } Z_1 = \sqrt{2} \operatorname{cis} \frac{\pi}{6}, Z_2 = \sqrt{2} \operatorname{cis} \frac{7\pi}{6}$$

(b) (i) Let $Z = \cos \theta + i \sin \theta$ be a root of unity

$$\text{So } (\cos \theta + i \sin \theta)^5 = 1$$

$$\cos 5\theta + i \sin 5\theta = 1$$

$$\cos 5\theta = 1$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$\text{Roots are: } Z_1 = 1, Z_2 = \operatorname{cis} \left(-\frac{2\pi}{5} \right), Z_3 = \operatorname{cis} \left(-\frac{4\pi}{5} \right),$$

$$Z_4 = \operatorname{cis} \frac{2\pi}{5}, Z_5 = \operatorname{cis} \frac{4\pi}{5}$$

(ii) Sum of roots is $Z_1 + (Z_2 + Z_4) + (Z_3 + Z_5)$ - Pairing conjugates

$$1 + 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = 0 \text{ (From } Z^5 - 1 = 0)$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$